



GMR-water: a water level retrieval software for multi-GNSS, multi-frequency and multi-observation based on GNSS-MR

Fei Yang¹ · Ruixian Hao¹

Received: 9 April 2025 / Accepted: 10 June 2026

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026

Abstract

Monitoring coastal sea levels is crucial for assessing the impacts of climate change and predicting flood risks. In this context, GNSS-Multipath Reflectometry (GNSS-MR) has emerged as a promising technique, offering an efficient and stable method for sea level retrieval. To fully realize the ability of various GNSS observations in GNSS-MR for the water level inversion, the GMR-Water, an open-source MATLAB-based software, has been developed which is capable of efficiently using the multi-observations including Signal-to-Noise Ratio (SNR), carrier phase, and pseudo range. Moreover, the GMR-Water supports the integration of multi-GNSS, multi-frequency, and multi-observation data, employing advanced error correction and combination algorithms to ensure high-precision retrieval of water levels. The software is designed with a user-friendly interface, equipped with batch processing capabilities, and it offers a comprehensive set of visualization tools to enhance the interpretation of results. We validated the performance of GMR-Water using observational data from the AT01 GNSS station, where the software was used to process the GNSS observation and analyze the water level measurement. The results demonstrate that GMR-Water can effectively retrieve water levels with high accuracy, achieving consistent and reliable results across various GNSS observation types. The performance of GMR-Water proves its potential as a valuable tool for precise coastal sea level monitoring, contributing to the further development and application of GNSS-MR water level monitoring technology.

Keywords GNSS-MR · Water level retrieval · Software · Multiple methods

Introduction

With the continuous development of global navigation satellite systems (GNSS) technology, its immense potential in environmental monitoring has become increasingly evident (Yang et al. 2023). Tens of thousands of GNSS observation stations have been established worldwide, including many located along coastlines (Li et al. 2024). While the multipath error caused by low elevation angle reflections off the water surface is generally considered to negatively impact positioning accuracy, recent studies have shown that the time delay characteristics between reflected and direct signals can be effectively used for tide level monitoring.

This technique, known as GNSS multipath reflectometry (GNSS-MR), offers the potential for existing GNSS networks to efficiently monitor global coastal tides and reservoir water levels, presenting broad application prospects. The comprehensive use of these systems can significantly enhance the temporal resolution of water level retrieval. However, differences in satellites number and frequency bands result in varying accuracy and resolution in water level retrieval. Therefore, current research focuses on evaluating the retrieval accuracy of each frequency band within different GNSS systems and integrating results across these systems. Furthermore, while traditional GNSS-MR water level retrieval relies primarily on SNR, other GNSS observations—such as carrier phase and pseudo range—also show potential for water level retrieval. This expands the application scenarios of GNSS-MR, particularly in situations where SNR data is unavailable.

The classical GNSS-MR water level retrieval technique analyzes the dominant frequency of the SNR residual sequence to estimate water levels (Larson et al. 2013). To

✉ Fei Yang
yangfei@cumtb.edu.cn

¹ College of Geoscience and Surveying Engineering, China University of Mining and Technology-Beijing, Beijing 100083, China

overcome its limitations, Strandberg et al. (2016) introduced an inverse modeling approach, treating reflection height as a model parameter and using B-spline curves with least-squares fitting for improved accuracy. The classical water level retrieval method is unavailable in the absence of SNR data. To address this, Ozeki and Heki (2012) proposed using the L4 dual-frequency combination from GPS observations for snow depth retrieval. Yu et al. (2015) extended this to GPS L1, L2, and L5 triple-frequency carrier phase combinations, achieving higher accuracy due to the low noise of phase measurements and mitigation of tropospheric and ionospheric delays. Zhang et al. (2022) proposed a method for water level retrieval using the CMC (code-minus-carrier phase) combination. Jin et al. (2017) demonstrated that water level retrieved based SNR, triple-frequency carrier phase, and pseudo range combinations are consistent with tide gauge reference, proving the capacity of carrier phase and pseudo range for accurate water level retrieval. As GNSS-MR technology evolves, corrections for errors such as tropospheric delays and sea surface dynamics enhance the accuracy of multi-frequency retrieval (Rousset et al. 2015; Williams and Nievinski 2017; Larson et al. 2017; Santamaría-Gómez and Watson 2017). Consequently, combining multi-frequency GNSS data is increasingly important. Wang et al. (2019) proposed a robust regression strategy using retrieval results within one-hour windows to produce hourly sea surface height sequences. Similarly, Purnell et al. (2020) used B-spline curves for uniformly sampled sequences derived from SNR inversions.

Environmental parameter inversion via GNSS-MR continues to advance, supported by various open-source tools. For example, Nievinski and Larson (2014) developed a GNSS multipath simulator, Martín et al. (2020) created a Python-based GNSS-IR tool, and Zhang et al. (2021) introduced GiRsnow for GNSS-IR-based snow depth retrieval. Additionally, Martín et al. (2022) developed software for converting GPS SNR phase data into volumetric water content, and Altuntas and Tunalioglu (2022) created GIRAS for GNSS-IR data analysis. Larson (2024) developed Gnsrefl, a Python tool for multi-environment parameter retrieval. Although the current software has made significant progress in retrieving snow depth, volumetric water, and soil moisture, etc., there are still limitations in the development of water level retrieval software. Traditional GNSS-MR techniques rely solely on SNR values, with no existing software utilizing carrier phase or pseudo range for retrieval. Additionally, inverse modeling method—a highly promising retrieval technique—has yet to be implemented in any available software. Furthermore, most existing methods focus only on single-frequency data and lack robust multi-frequency combination approaches, making them insufficient for generating uniformly sampled data.

To address these challenges, we developed GMR-Water, a MATLAB-based software supporting water level retrieval with multi-GNSS (GPS, GLONASS, Galileo, BDS), multi-frequency and multi-observation types (SNR, carrier phase, pseudo range). GMR-Water incorporates advanced error correction methods and robust combination estimation strategies, ensuring high-precision output water level.

Principle of water level retrieval

GNSS receivers are designed to capture electromagnetic signals transmitted by satellites, and GNSS antennas are primarily sensitive to direct signals from these satellites. However, in practical scenarios, the antenna also receives reflected signals from surrounding objects in addition to the direct signals. These direct and reflected signals interfere and combine at the antenna, meaning the received signal is a composite of both. Due to the reflection process, the intensity of reflected signals is weaker than that of direct signals. Nevertheless, these reflected signals contain valuable information about the reflecting surface and the reflection path. By isolating the portion of the GNSS signal affected by multipath errors, it becomes possible to infer the characteristics of the reflecting surface and path. This principle underlies the GNSS-MR for water level retrieval (as shown in Fig. 1), where the reflection height, as a key characteristic of the reflection path, can be accurately determined.

GNSS observations typically include carrier phase, pseudo range, and SNR, which are affected by multipath signals and can generally be expressed as a combination of the original observation and the multipath component. The GNSS-MR technology derives the water level information by analyzing the multipath component, so the accuracy of the multipath information is crucial in the entire process. In the developed GMR-Water software, five models were provided for isolating multipath information, as detailed in Tables 1 and 2.

As shown in Fig. 1, the analysis of multipath information typically involves spectral analysis to identify the dominant frequency of the signal, which can be inverted to the reflection height. This process is commonly performed using the Lomb–Scargle Periodogram (LSP) method. Once the reflection height is obtained, applying error corrections yields the final water level height. It should be noted that the SNR inverse modeling method does not follow these principles. Instead, it treats the reflection height as a parameter within the SNR residual model, which is determined using the least-squares estimation method.

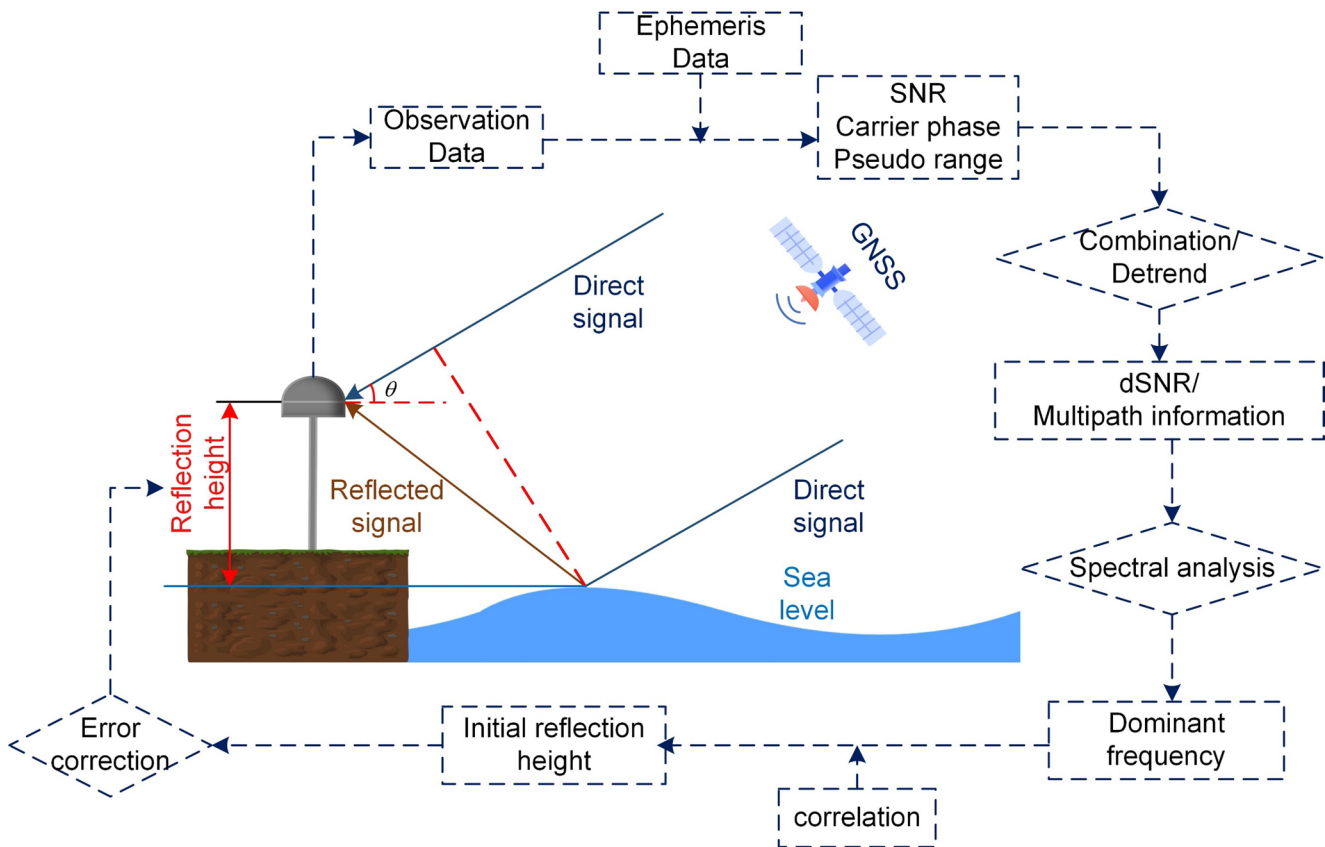


Fig. 1 Geometry and principles of GNSS water level retrieval

Table 1 GNSS observation expressions and main formulas of multipath model

Observation	Multipath models
$SNR = \sqrt{A_d^2 + A_r^2 + 2A_dA_r \cos \Delta\varphi}$	$\delta SNR = 2A_dA_r \cos \Delta\varphi$ (Larson et al. 2013)
	$\delta SNR = A_e E \cos \left(\frac{4\pi H \sin \epsilon}{\lambda} + \varphi_0 \right)$ (Strandberg et al. 2016)
$L_i = \rho - I(f_i) + T + M_{Li} - \lambda_i N_i$	$M_{L1,2,3} = (\lambda_3^2 (L_1 - L_2) + \lambda_1^2 (L_2 - L_3) + \lambda_2^2 (L_3 - L_1)) / 1m^2$ (Yu et al. 2015)
	$M_{L1,2} = L_1 - L_2 = I(f_2) - I(f_1) + M_{L1} - M_{L2} + \lambda_2 N_2 - \lambda_1 N_1$ (Ozeki and Heki 2012)
	$M_i = C_i - L_i = 2I(f_i) + M_{Ci} - M_{Li} + \lambda_i N_i$ (Zhang et al. 2022)
$C_i = \rho + I(f_i) + T + M_{Ci}$	$M_{C1,2,3} = (\lambda_3^2 (C_1 - C_2) + \lambda_1^2 (C_2 - C_3) + \lambda_2^2 (C_3 - C_1)) / 1m^2$
	$M_{C1,2} = C_1 - C_2 = I(f_2) - I(f_1) + M_{C1} - M_{C2}$

GMR-water software

With the rapid development of GNSS-MR water level retrieval technology, many retrieval errors have been corrected, and more advanced data processing strategies have been introduced. The accuracy and reliability of water level retrieval results using traditional strategies have been

significantly improved. Therefore, GMR-Water integrates these advancements and adds some new modules to form a comprehensive and advanced water level retrieval software based on multi-GNSS, multi-frequency and multi-observation. More specifically, GMR-Water provides a one-stop platform, from GNSS observation to water level, further advancing the development and application of GNSS-MR

Table 2 Definitions of variables for the formulas shown in Table 1

Variables	Meanings
A_d	The amplitudes of the direct signal
A_r	The amplitudes of the reflected signal
$\Delta\varphi$	The phase difference between the direct signal and the reflected signal
A_e	The amplitude independent of the elevation angle
E	The attenuation coefficient of surface
λ	Carrier wavelength
H	Reflection height
e	Satellite elevation angle
φ_0	The phase contribution of the antenna pattern and electromagnetic properties of the reflecting surface
L_i	Carrier
C_i	Pseudocode
ρ	The geometric range between the satellite and receiver
M	The multipath error
N	Integer ambiguity
I	The ionospheric delay
T	The tropospheric delay

water level inversion technology. The main features of GMR-Water compared to only a few open-source software include:

- Multi-GNSS: water level retrieval based on GPS, Galileo, GLONASS, and BDS.
- Multi-Frequency: support for all frequency bands.
- Multi-Observation: water level retrieval based on SNR, Carrier phase and pseudo range.
- Advanced error correction techniques (tropospheric correction, dynamic tide level correction etc.) are applied for high-precision water level retrieval.
- Efficient data processing capability through parallel batch processing.
- Innovative methods (robust regression strategy and the B-spline method) for the combination of multi-frequency results.
- Comprehensive data visualization tools and a user-friendly interactive interface.

The software is divided into four main parts, with its structure and operational flow illustrated in Fig. 2.

The first part focuses on preparing the files necessary for water level retrieval. Three files are required: the RINEX file, which records various observations (SNR values, carrier phase, and pseudo-distance); the station file, containing information such as the geographical position and effective elevation and azimuth angles range; and the ephemeris file, which provides the position of satellites. Moreover, if the users want to evaluate the accuracy of the water level, a reference tide file from a tide station must also be provided.

The second part involves extracting the SNR, carrier phase, and pseudo range information needed for retrieval from the input files. For GNSS-MR water level retrieval, only observation data of specific azimuth and elevation are needed, so the data needs to be screened in this part. Firstly, the coordinate of the station provided in the station file and the coordinate of the satellite at each time provided in the ephemeris file are used to calculate the elevation and azimuth angle corresponding to the observed values. Further, the effective elevation and azimuth angle range provided in the station file are used to screen the observed values, and the final output is observation data suitable for water level retrieval, including SNR, carrier phase, and pseudo range.

The third part focuses on water level retrieval. GMR-Water provides five water level retrieval methods based on three types of observations. For SNR observations, water level retrieval can be performed using the classical spectral analysis or the inverse modeling method. For carrier phase and pseudo range observations, water level retrieval can be achieved either through dual-frequency/tri-frequency combinations of a single type of observation or through a combination of single-frequency carrier phase and pseudo range data. Detailed mathematical principles are provided in Sects. "[Principle of water level retrieval](#)" of the user manual. Using water level retrieval models, GMR-Water can convert GNSS observation data into water level data and output it in a standardized data format. Refer to Sect. "[GMR-Water software](#)" of the user manual for detailed software operation instructions.

The water level information inverted from GNSS observations has been obtained through the previous three parts. In the final section, GMR-Water provides an analysis module that performs two main functions: combining water level information retrieved from multi-frequency GNSS data and evaluating the accuracy of the retrieved water levels using tidal reference data. GMR-Water offers two combination methods: a robust regression strategy and B-spline curve estimation. Through combination, higher-accuracy water level data with uniform sampling intervals can be obtained. In the accuracy evaluation part, indicators such as RMSE, Bias, and the number of daily inversion points are calculated, and comprehensive visualization results are generated.

Experiments and results

In this section, we utilize GMR-Water to conduct water level retrieval at the AT01 station for the period from February 1 to February 28, 2023. The retrieval results are based on five methods, along with an assessment of combination techniques will be presented, and all the pictures are exported from the software.

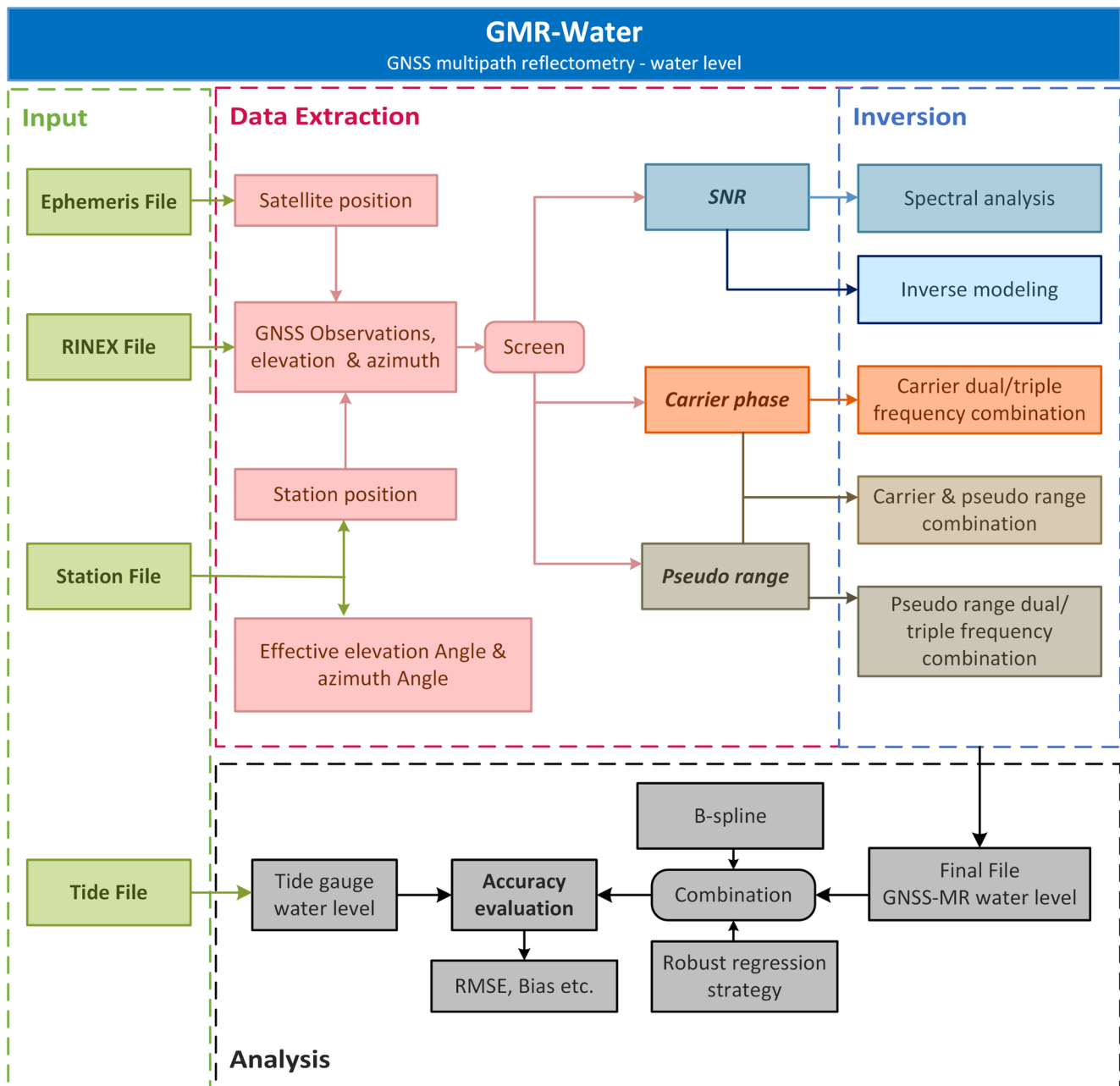


Fig. 2 The structure and operation flow of GMR-Water

Experimental station

The AT01 Observatory (63.48405°N, 162.00640°W) is situated in St. Michael, Alaska, United States, and is part of the EarthScope Plate Boundary Observatory (PBO) research initiative. The vertical distance from the receiver antenna phase center to sea level is approximately 12 m. The station supports signal reception from multiple GNSS, including GPS (L1, L2, L5), GLONASS (G1, G2), Galileo (E1, E5a, E5b, E5, E6), and BeiDou (B1, B1-2, B2a, B2b, B3). Additionally, the NOAA-operated Tide Station 9,468,333

(63°52.3'N, 160°47.1'W) is located 74 km from the AT01 Observatory, with a 6-min sampling interval. To facilitate the accuracy assessment, the tide gauge data were interpolated to the GNSS-MR retrieval time points using spline interpolation. The spatial relationship between Tide Station 9,468,333 and the AT01 GNSS station is depicted in Fig. 3.

The reflected signals primarily originate from the first Fresnel zone. Therefore, understanding the extent of the first Fresnel zone is crucial for determining whether the reflected signals originate from the sea surface, which in turn aids in defining appropriate azimuth and elevation angle ranges.

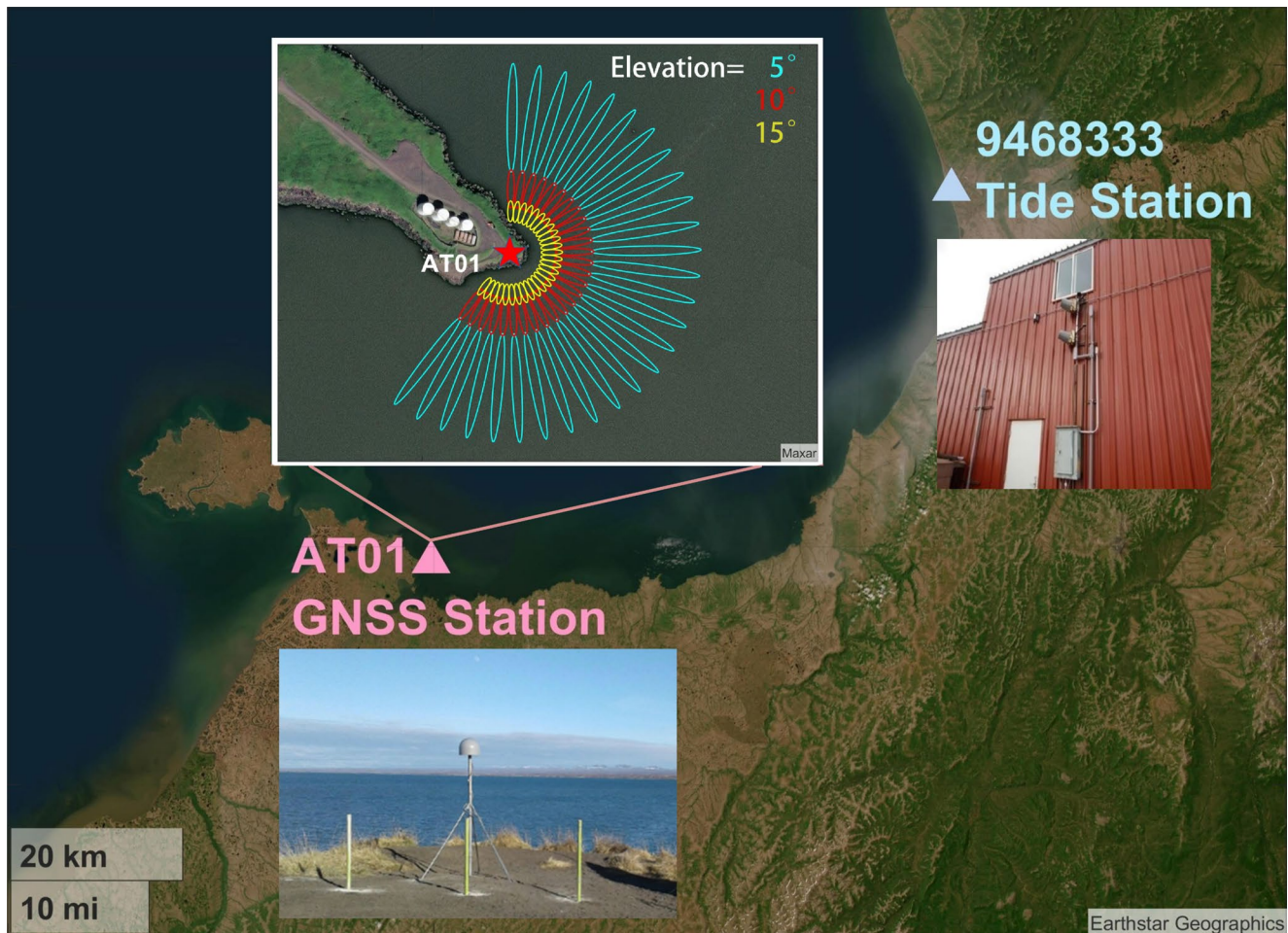


Fig. 3 Geographical location diagram of the AT01 GNSS station and the 9,468,333 tide station, along with a real scene depiction and the first Fresnel zone

For the AT01 station, the azimuth is restricted to the range of 0° to 220° , and the elevation angle is constrained between 5° and 15° . Figure 3 illustrates the first Fresnel zone at elevation angles of 5° , 10° , and 15° . GMR-Water supports the output of Fresnel reflection zone information for each reflection and provides visualization tools to ensure that reflection occurs within appropriate area.

Retrieval based on SNR

The water level retrieval was first conducted using the SNR spectral analysis method in the developed GMR-Water software. Figure 4 presents the time series of the retrieval results for different systems and different frequencies, along with the RMSE and the daily average number of inversion. Benefiting from the advanced inversion model and quality control methods, GMR-Water demonstrates competent performance across all systems and frequency bands. The RMSE for water level retrieval at all frequency bands are below 20 cm. Among these, the highest retrieval accuracy is

achieved at the GPS S1L frequency point, with an RMSE of 17.3 cm, while the lowest retrieval accuracy is observed at the BDS S1P frequency band, with an RMSE of 19.681 cm. Additionally, due to the larger number of satellites and frequency bands available in BDS, it achieved the maximum daily average number of inversion, exceeding 263 per day. In contrast, Galileo had the minimum value, with only 180 per day. Overall, GMR-Water performs excellently in water level retrieval across all GNSS and frequency bands, delivering high-precision and reliable water level retrieval results.

In addition, GMR-Water offers two combination strategies for multi-frequency results. Figure 5 (top) shows the results of the robust regression strategy and B-spline curve estimation. The gray points represent the raw retrieval scatter, while the red curve represents the combined retrieval results. Through these combination strategies, GMR-Water generates water level sequences with equal sampling intervals, improving the applicability of GNSS-MR water level retrieval results. Additionally, by integrating multi-frequency results, more accurate

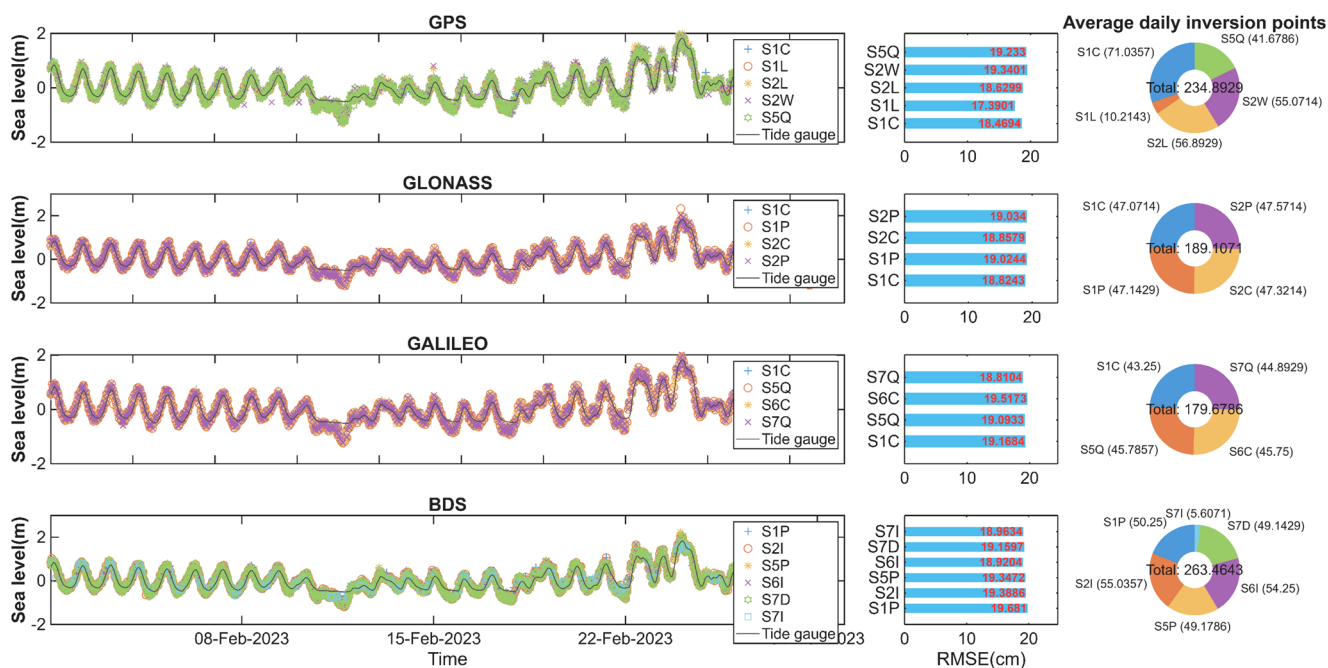


Fig. 4 Retrieval results for the AT01 station from February 1 to February 28, 2023, including multi-system, multi-frequency time series, RMSE at each frequency, and the number of average daily inversion points

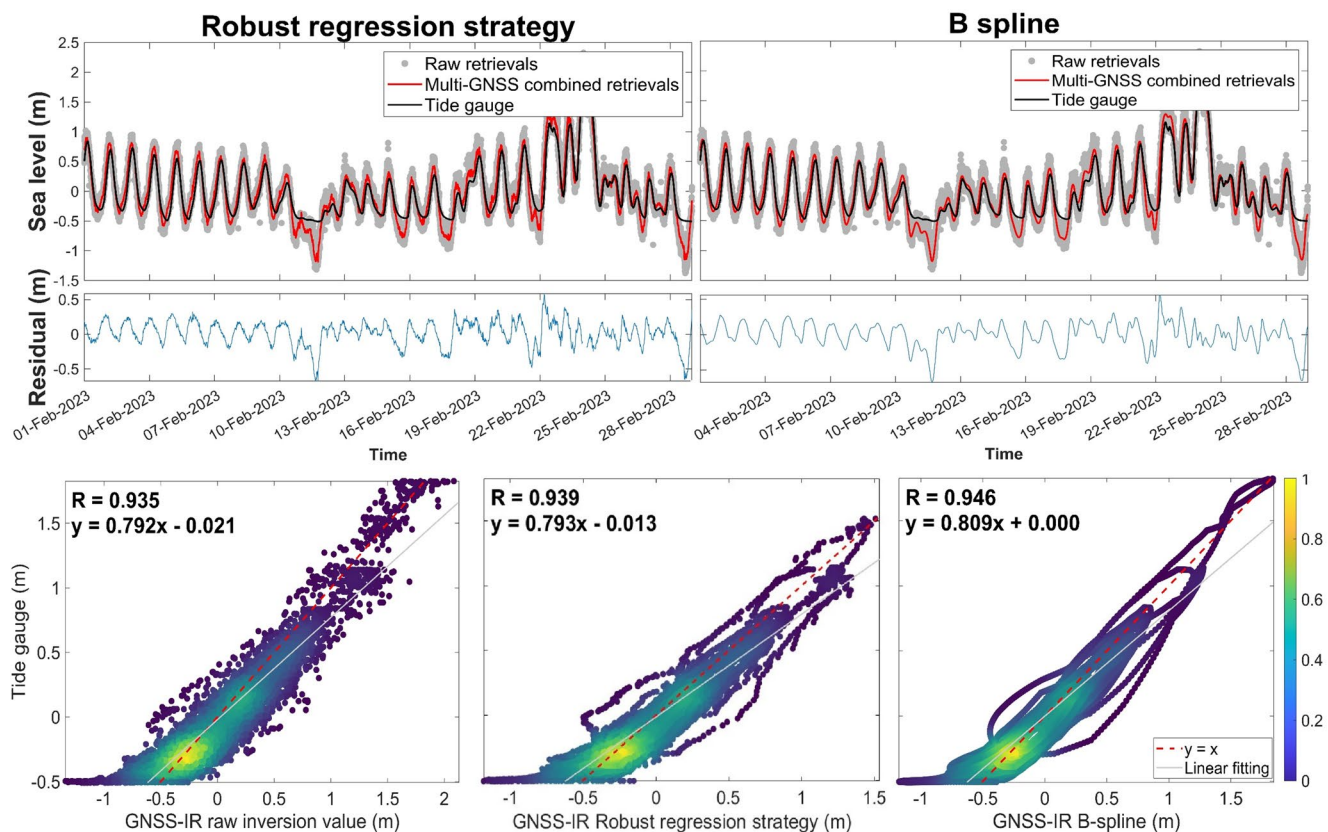


Fig. 5 Time series and residual for two combination strategies (top), along with the scatter density plot (bottom)

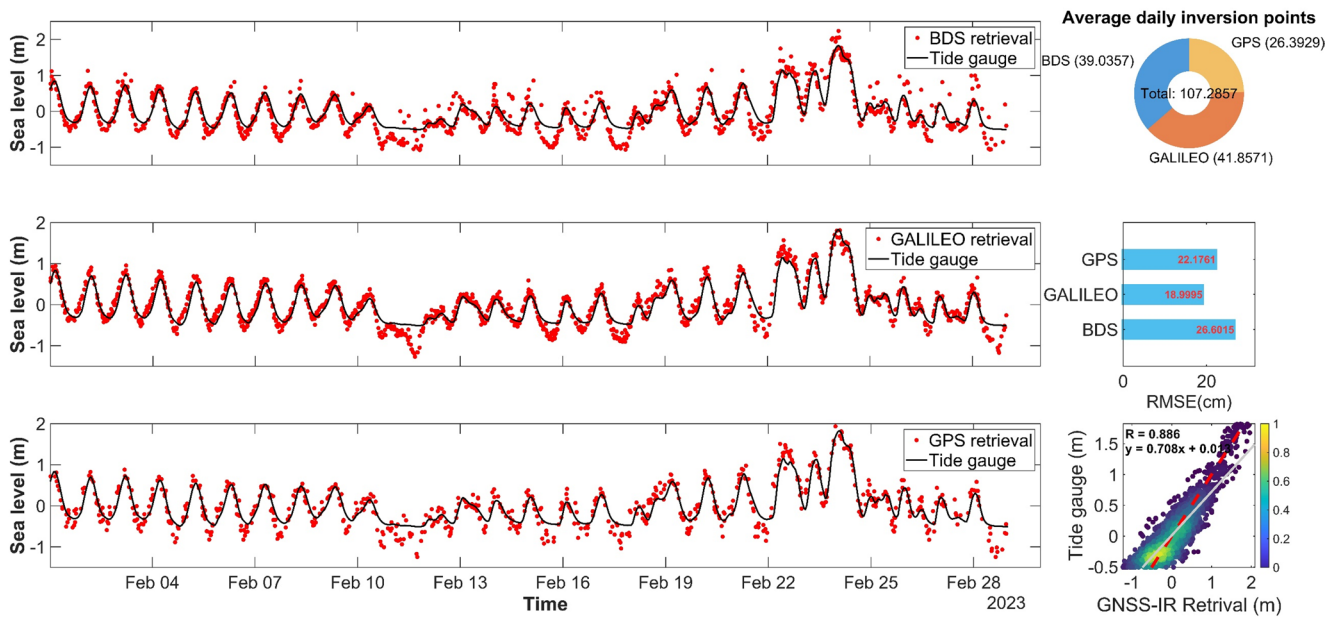


Fig. 6 Water level retrieval results using the triple-frequency carrier phase combination method

retrieval results were obtained. Figure 5 (bottom) presents scatter density plots of the raw GNSS-MR retrieval results, robust regression strategy combination results, B-spline curve estimation results, and reference values from the tide gauge station. In comparison with the original retrieval results, both combination strategies yielded increased correlation coefficients of 0.939 and 0.946, up from 0.935. However, since this method is based on curve fitting, the results are determined by the characteristics of the B-spline curve and may inherently have some unreasonable aspects. Therefore, the robustness of the results obtained using the robust regression strategy is more reliable.

Retrieval based on carrier phase and pseudo range

In addition to SNR, both carrier phase and pseudo range also possess the capability for water level retrieval. GMR-Water provides various water level retrieval methods based on these two types of observations. Figure 6 presents the results of water level retrieval using a three-frequency carrier combination from the GPS, Galileo, and BDS systems. Specifically, GPS uses the L1 (1575.42 MHz), L2 (1227.60 MHz), and L5 (1176.45 MHz) frequency bands for the combination, Galileo utilizes the E1 (1575.42 MHz), E5a (1176.45 MHz), and E5b (1207.14 MHz) frequencies, and BDS uses the B1 (1575.42 MHz), B3 (1268.52 MHz), and B2a (1176.45 MHz) bands. It is evident that GMR-Water demonstrates competent water level retrieval performance based on carrier phase observations, with RMSE of 22.17 cm, 19.00 cm, and 26.60 cm for the three systems, respectively. The correlation coefficient between the water

levels retrieved from the carrier phase and the reference values from tide gauge stations is 0.886, indicating a significant linear relationship. The daily average number of retrieval points for the three systems is 26.39, 41.86, and 39.03, respectively. Compared to SNR-based water level retrieval, carrier phase retrieval yields lower accuracy and fewer retrieval points. Thus, further optimization of the water level retrieval model based on carrier phase is still required.

In addition, GMR-Water supports SNR inverse modeling, dual/triple-frequency pseudo range combination, and single-frequency carrier phase and pseudo range combination for water level retrieval. For more experimental results, please refer to Sect. 7 (Examples) of the user manual.

Conclusions

The GMR-Water, an open-sourced GNSS-MR-based water level retrieval software, which is fully utilize the multi-GNSS, multi-frequency and multi-observation to construct different inversion models. The software is also equipped with robust combination techniques, enabling seamless processing from GNSS observations to the water level information with uniform sampling intervals. It features powerful data processing capabilities alongside an intuitive user interface. The experiment conducted in AT01 site using the GMR-Water software showed the functionality and features of the developed software, and demonstrated the software can achieve high-precision water level results using different models and GNSS observations. We are committed to continuously updating and improving the software.

Author contributions F.Y. conceived and designed the study, F.Y. and R.H. developed the software, R.H. conduct the experiment and formal analysis, R.H. wrote the main manuscript, F.Y. reviewed the manuscript.

Funding This research was founded by the National Natural Science Foundation of China (42204022), the Fundamental Research Funds for the Central Universities (2024ZKPYDC02, 2024CXCY010), Open Fund of Key Laboratory of Beidou Navigation and Positioning, Department of Natural Resources of Jiangsu Province (KLBDS-G202502), the Laboratory of Southeast Coast Marine Information Intelligent Perception and Application, MNR (24102), the Open Research Fund of State Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University (23P02).

Data availability The GNSS observation data of AT01 site can be downloaded from the Plate Boundary Observatory (PBO) (<http://pbo.unavco.org>) and the tide gauge data are available through the National Oceanic and Atmospheric Administration (NOAA) for the tide gauge data (<https://www.tidesandcurrents.noaa.gov>).

Code availability The software and manual are provided at <https://github.com/GRseRG-CUMTB/GMR-Water>.

Declarations

Competing interests The authors declare no competing interests.

References

- Altuntas C, Tunalioglu N (2022) GIRAS: an open-source MATLAB-based software for GNSS-IR analysis. *GPS Solut* 26:16. <https://doi.org/10.1007/s10291-021-01201-3>
- Jin SG, Qian XD, Wu X (2017) Sea level change from BeiDou Navigation Satellite System-Reflectometry (BDS-R): First results and evaluation. *Glob Planet Change* 149:20–25. <https://doi.org/10.1016/j.gloplacha.2016.12.010>
- Larson KM (2024) Gnsrefl: an open source software package in python for GNSS interferometric reflectometry applications. *GPS Solut* 28:165. <https://doi.org/10.1007/s10291-024-01694-8>
- Larson KM, Löfgren JS, Haas R (2013) Coastal sea level measurements using a single geodetic GPS receiver. *Adv Space Res* 51(8):1301–1310. <https://doi.org/10.1016/j.asr.2012.04.017>
- Larson KM, Ray RD, Williams SDP (2017) A 10-Year comparison of water levels measured with a geodetic GPS receiver versus a conventional tide gauge. *J Atmos Ocean Technol* 34:295–307. <https://doi.org/10.1175/JTECH-D-16-0101.1>
- Li J, Yang F et al (2024) Unraveling the accuracy enigma: Investigating ZTD data precision in TUW-VMF3 and GFZ-VMF3 products using a comprehensive global GPS dataset. *IEEE Trans Geosci Remote Sens* 62:1–1. <https://doi.org/10.1109/TGRS.2024.3385228>
- Martín A, Luján R, Anquela AB (2020) Python software tools for GNSS interferometric reflectometry (GNSS-IR). *GPS Solut* 24:94. <https://doi.org/10.1007/s10291-020-01010-0>
- Martín A, Anquela AB, Ibáñez S et al (2022) Python software to transform GPS SNR wave phases to volumetric water content. *GPS Solut* 26:7. <https://doi.org/10.1007/s10291-021-01190-3>
- Nievinski FG, Larson KM (2014) An open source GPS multipath simulator in Matlab/Octave. *GPS Solut* 18:473–481. <https://doi.org/10.1007/s10291-014-0370-z>
- Ozeki M, Heki K (2012) GPS snow depth meter with geometry-free linear combinations of carrier phases. *J Geod* 86(3):209–219. <https://doi.org/10.1007/s00190-011-0511-x>
- Purnell D, Gomez N, Chan NH et al (2020) Quantifying the uncertainty in ground-based GNSS-Reflectometry sea level measurements. *IEEE J-STARS* 13:4419–4428. <https://doi.org/10.1109/JSTARS.2020.3010413>
- Roussel N, Ramillien G, Frappart F, Darrozes G et al (2015) Sea level monitoring and sea state estimate using a single geodetic receiver. *Remote Sens Environ* 171:261–277. <https://doi.org/10.1016/j.rse.2015.10.011>
- Santamaria-Gómez A, Watson C (2017) Remote leveling of tide gauges using GNSS reflectometry: case study at Spring Bay, Australia. *GPS Solut* 21:451–459. <https://doi.org/10.1007/s10291-016-0537-x>
- Strandberg J, Hobiger T, Haas R (2016) Improving GNSS-R sea level determination through inverse modeling of SNR data. *Radio Sci* 51:1286–1296. <https://doi.org/10.1002/2016RS006057>
- Wang XL, He XF, Zhang Q (2019) Evaluation and combination of quad-constellation multi-GNSS multipath reflectometry applied to sea level retrieval. *Remote Sens Environ* 231(2):111229. <https://doi.org/10.1016/j.rse.2019.111229>
- Williams SDP, Nievinski FG (2017) Tropospheric delays in ground-based GNSS multipath reflectometry—experimental evidence from coastal sites. *J Geophys Res Solid Earth* 122:2310–2327. <https://doi.org/10.1002/2016JB013612>
- Yang F, Sun Y, Meng X et al (2023) Assessment of tomographic window and sampling rate effects on GNSS water vapor tomography. *Satell Navig* 4:7. <https://doi.org/10.1186/s43020-023-00096-4>
- Yu K, Ban W, Zhang X, Yu X (2015) Snow depth estimation based on multipath phase combination of GPS triple-frequency signals. *IEEE Trans Geosci Remote Sens* 53(9):5100–5109. <https://doi.org/10.1109/TGRS.2015.2417214>
- Zhang S, Peng J, Zhang C et al (2021) GiRsnow: an open-source software for snow depth retrievals using GNSS interferometric reflectometry. *GPS Solut* 25:55. <https://doi.org/10.1007/s10291-021-01096-0>
- Zhang Z, Guo F, Zhang X et al (2022) First result of GNSS-R-based sea level retrieval with CMC and its combination with the SNR method. *GPS Solut* 26:20. <https://doi.org/10.1007/s10291-021-01208-w>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.



Fei Yang received the Ph.D. degree in geodesy and geomatics from Wuhan University, Wuhan, China, in 2020. He is currently an associate professor at the China University of Mining and Technology-Beijing. His main research interests include GNSS meteorology and GNSS-IR.



Ruixian Hao is currently working toward the M.Sc. degree of Geodesy and Geomatics in China University of Mining and Technology-Beijing. His main research interest is GNSS-IR.